Red Team Critique of Blue Team Version Two

Red Team review · November 4, 2026

Blue Team version two is a clear improvement in vocabulary and a modest improvement in substance. It names the right problems, but for most of them it puts a technique name where the mechanism should be. Several fixes are wrong in detail, and a few reintroduce the faults they were meant to remove.

Critical faults

1. The capacity arithmetic is wrong, and there are still no real numbers. Version two says capacity multiplied by recall equals actionable cases. Recall is the share of all abusers the system catches, and it does not scale analyst capacity. Precision is what connects the two: confirmed cases equal completed reviews multiplied by precision. Here is an illustration. With one million merchants and 0.2 percent abusing accounts, there are 2,000 abusers. If analysts complete 1,000 reviews per week, one quarter gives 13,000 reviews. To confirm 1,800 abusers (ninety percent recall), the queue needs about 14 percent precision across everything reviewed. Version two says targets must exist but states none, and its threshold is still the placeholder "X".

2. The pre-review gate does not reduce reviewer workload, and it invites gaming. A rule that raises an immediate flag adds cases to the queue. Reducing the queue requires automatically clearing low-risk merchants, which costs recall, and version two never discusses that trade. A fixed velocity threshold within five minutes also fires on lunch rushes, festival stalls, and ticket counters. Abusers will learn to stay just below it, which is the gaming problem from Red Team fault nine.

3. Champion and challenger retraining still learns from its own decisions. Labels come only from merchants the system flagged and an analyst confirmed. Abusers the system missed are never labelled, so each weekly retraining makes the model better at finding what it already finds. The fix needs randomly sampled audits of unflagged merchants and a protected holdout set. Investigations also take weeks, so weekly retraining uses labels that are incomplete or stale.

4. Smoothing and Kalman filtering follow the drift instead of detecting it. Both techniques are built to track a changing level. A merchant whose volume climbs gradually toward illicit levels is tracked, and the tracked value becomes the new normal. That is the

exact failure the Red Team described. Detecting slow drift needs a cumulative change test measured against the fixed anchors. Version two names both ideas but never connects them.

5. The anchors and cohorts can be chosen or contaminated by the abuser. The onboarding profile is declared by the merchant, who can pick a broad category with high natural variation. Cohorts are built from that declared category. In areas where abuse is widespread, the cohort absorbs it and treats it as normal. Small cohorts have too little data for reliable comparison, and new merchants have no history at all. Honest businesses that change direction are also punished, because version two offers no process for updating the declared profile.

Major faults

6. The account-control signals flag honest merchants and miss rented accounts.

- Owners buy new phones, travel, use mobile networks whose addresses are shared among many subscribers, and use virtual private networks for ordinary privacy. A family member or employee operating the till is legitimate delegation, not rental.
- Many sole proprietors settle to their own personal savings accounts. A rule that flags non-business payout accounts flags a large share of honest small shops.
- Relationship mapping through shared addresses and registered directors cannot find a rented account, because the renter deliberately does not appear in registration records. The renter is visible only through behavior: who pays the account, who the account pays, and how it is operated.

7. The pass-through ratio is defined backwards. Total inflow divided by total outflow being high means money is accumulating. Pass-through means outflow nearly equals inflow, so the ratio sits close to one and the balance stays near zero. Marketplaces, landlords, travel agents, wholesalers, and platforms that pay gig workers share that shape, so the feature must be combined with payer diversity and cohort comparison.

8. The betting features describe the wrong direction and too narrow a slice.

- The signature of betting collection is many distinct payers sending repeatedly to one account, in round stake-like amounts, often from payers far from the merchant's location. "Many identical amounts to one recipient" is a different pattern. Identical amounts also describe fixed-price tea stalls, tuition fees, subscriptions, and parking.

- Winnings are more likely to be many small or moderate payments tied in time to payments from the same people than "large, one-time outflows". Payouts may also leave through cash or other channels and never appear in the account.
- The event calendar covers match-timed sports only. Casino-style games, card games, lotteries, and fantasy competitions run all day. The stadium vendor and sports bar overlap also remains unresolved.

9. The privacy technique does not create a legal basis or governance. Federated learning is a training method and not a linkage method. The relevant tool, private set intersection, is not named. Version two then says identifiers are shared anyway, which is the disclosure the technique was meant to avoid. It also leaves open who operates the consortium, what evidence is required before a score is shared, how a wrong score is corrected, and who is liable. Scores from different providers are calibrated differently and cannot be compared directly. Without an appeal path, one provider's mistake becomes industry-wide exclusion. The footprint adjustment is also circular, because it gives leniency to high-volume merchants and so rewards the very volume under suspicion.

10. The legal triage and the explanation requirement contradict each other.

- Suspicious transaction reporting rules in many countries forbid warning a customer that their activity is being reported. Mandatory reason codes delivered to the merchant can conflict with that.
- Even without that conflict, a reason such as "login network address mismatch" tells an abuser exactly which signal to imitate. The design needs two levels of explanation: a general one for the merchant and a detailed one for reviewers.
- Version two says the system must not decide legality, but a label of "betting risk" is already a legal characterization. Routing every flagged case to legal review also contradicts the capacity limits it stated earlier. The appeal path has no actual time limit and says nothing about funds held during an appeal.

11. Several Red Team faults receive no response, yet the summary claims all are addressed. Gaming of the intervention tiers (nine), scale and cost estimates (twelve), and insider and process risk (thirteen) are not answered. Seasonality (eleven) is only partly addressed. The three-zone intervention ladder from version one also disappears. The action layer is a single paragraph with no thresholds, no time limits on holds, and no list of restrictive actions.

What Blue Team version three should fix

Fault	Required fix
Capacity arithmetic	Provide a worked example linking prevalence, precision, and review capacity, with stated targets
Pre-review gate	Add a clearing path for low-risk merchants and randomize thresholds so they cannot be learned
Label feedback loop	Audit a random sample of unflagged merchants and keep a protected holdout set
Drift detection	Use cumulative change tests against fixed anchors, not smoothing alone
Anchors and cohorts	Use cohorts built from observed behavior, and allow re-declaration of the business profile
Account control	Use counterparty-based signals, and treat family and employee delegation as normal
Pass-through ratio	Define it as outflow close to inflow with a near-zero balance, combined with payer diversity
Betting features	Focus on repeat payer concentration, stake-like amounts, payer distance, and all-day game types
Information sharing	Name the legal basis, governance, liability, and correction process
Explanations and reporting	Provide two levels of explanation and reconcile them with reporting secrecy rules
Missing faults	Answer tier gaming, scale cost, and insider risk directly

Bottom line

Version two replaced missing mechanisms with technique names. The strongest next version would lead with one worked quantitative example, add random audits to break the label loop, and build account-control signals from payers, payees, and operator behavior instead of registration records.